

Samuele TONDELLI

Computer Science and Engineering Student

 [linkedin.com/in/samuele-tondelli-681691356](https://www.linkedin.com/in/samuele-tondelli-681691356)

 [github.com/SamueleTondelli](https://github.com/SamueleTondelli)

 [Personal website](mailto:tondellisamuele1@gmail.com)

 [tondellisamuele1@gmail.com](mailto:tondellisamuele1@gmail.com)

“ SUMMARY

Final-year master’s student in Computer Science and Engineering focused on ML systems, GPU computing, and performance engineering. Currently optimizing GPU inference of a graph transformer for protein docking. Background in low-level systems programming, FPGA acceleration, and embedded software. Seeking internship and research opportunities.

🎓 EDUCATION

|                |                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - Present | Master in Computer Science and Engineering, <b>Politecnico di Milano</b>                                                                                                                                                 |
| 2021 - 2024    | Bachelor in Computer Engineering, <b>University of Modena and Reggio Emilia</b> . Thesis : "Optimization and synthesis on FPGA of the logarithm of softmax function for neuromorphic accelerators". <b>110 cum laude</b> |

☰ SKILLS

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| Languages           | C/C++, Python, Rust, Assembly, Go, Java                       |
| GPU & performance   | CUDA, SIMD, Triton, AVX-512, TensorRT, NSight sytems/compute  |
| ML frameworks       | PyTorch, Scikit-learn, Hydra, TensorFlow                      |
| Hardware & Embedded | Vitis HLS, STM32, Verilog, Xilinx Zynq FPGAs, Raspberry Pi    |
| Dev tools           | Linux, Git, Make, CMake, GCC, GDB, Valgrind, Vivado, ModelSim |

🧰 EXPERIENCE

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|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feb 2026<br>Sep 2025 | <b>Student Researcher   Genomic ML, NECSTLAB, POLITECNICO DI MILANO, Italy</b> <ul style="list-style-type: none"><li>➤ Conducting survey study on ML approaches for single nucleotide variant classification.</li><li>➤ Benchmarked 4 models (MLP, XGBoost, Logistic Reg, Random Forest) across 144 genomic datasets.</li><li>➤ Reduced training resource usage up to 10x while minimally impacting test performance.</li><li>➤ Analyzed pathogenicity prediction on variants of uncertain significance (90% consistency across 72 models).</li><li>➤ Paper accepted at IEEE EMBC 2026 (Paper No. 3391).</li></ul> <div>ML Genomics Python Hydra Scikit-learn</div> |
| Sep 2025<br>Mar 2025 | <b>Student Researcher   FPGA Deep Space, NECSTLAB, POLITECNICO DI MILANO, Italy</b> <ul style="list-style-type: none"><li>➤ Accelerated ECOS library for autonomous navigation of deep space satellites.</li><li>➤ Developed accelerator for sparse matrix computations (SpMV and LDLt kernels) using Vitis HLS and Vivado.</li><li>➤ Deployed IP on Kria KD240 and developed a C Linux driver, achieving 2x speedup over software.</li><li>➤ Competed in the 2025 edition of the AMD Open Hardware competition.</li></ul> <div>FPGA HLS Vitis Vivado Xilinx Embedded systems C</div>                                                                               |
| May 2024<br>Mar 2023 | <b>Software Engineer Intern   Formula SAE, MORE MODENA RACING, UNIMORE, Italy</b> <ul style="list-style-type: none"><li>➤ Co-led a team of 4 developing software for combustion and hybrid Formula SAE cars.</li><li>➤ Developed live, wireless telemetry (1 Mbps over 500m) using Raspberry Pi 5 and Grafana.</li><li>➤ Developed car dashboard software on STM32 controller reading CAN bus data.</li></ul> <div>Automotive C Raspberry Pi Live telemetry Grafana STM32 CAN</div>                                                                                                                                                                                 |

💻 PROJECTS

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| <b>GPU ACCELERATED GRAPH TRANSFORMER INFERENCE</b> <div>Master’s thesis   Politecnico di Milano, private repository</div> <div>Optimizing GPU inference of a graph transformer for protein docking, in active development.</div> <div>GPU CUDA Triton PyTorch TensorRT Graph Transformer NSight</div>                                                                                                             | 2026 |
| <b>GRADINO - OPTIMIZED C AUTOGRAD ENGINE</b> <div> <a href="https://github.com/SamueleTondelli/gradino">github.com/SamueleTondelli/gradino</a></div> <div>Autograd engine written in C optimized for CPU training with SIMD acceleration and custom allocators.</div> <div>C Deep learning SIMD AVX-512 Arena allocators</div> | 2026 |

🌐 LANGUAGES

|         |       |
|---------|-------|
| Italian | ●●●●● |
| English | ●●●●○ |

+ STRENGTHS

- Curiosity
- Motivation
- Passion